

Recent trends and advances in berry health benefits research.

Recent advances have been made in our scientific understanding of how berries promote human health and prevent chronic illnesses such as some cancers, heart disease, and neurodegenerative diseases. Cancer is rapidly overtaking heart disease as the number one killer disease in developed countries, and this phenomenon is coupled with a growing aging population and concomitant age-related diseases. Therefore, it is not surprising that consumers are turning toward foods with medicinal properties as promising dietary interventions for disease prevention and health maintenance. Among fruits, berries of all colors have emerged as champions with substantial research data supporting their abilities to positively affect multiple disease states. Apart from several essential dietary components found in berries, such as vitamins, minerals, and fiber, berries also contain numerous bioactives that provide health benefits that extend beyond basic nutrition. Berry bioactives encompass a wide diversity of phytochemicals (phytonutrients) ranging from fat-soluble/lipophilic to water-soluble/hydrophilic compounds. Recent research from laboratories across the globe has provided useful insights into the biological effects and underlying mechanisms of actions resulting from eating berries. The cluster of papers included here represents a cross section of topics discussed at the 2009 International Berry Health Benefits Symposium. Together, these papers provide valuable insight into recent research trends and advances made into evaluating the various health benefits that may result from the consumption of berries and their derived products.